

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Karanam Bhaskar Gnanesh(192324080)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

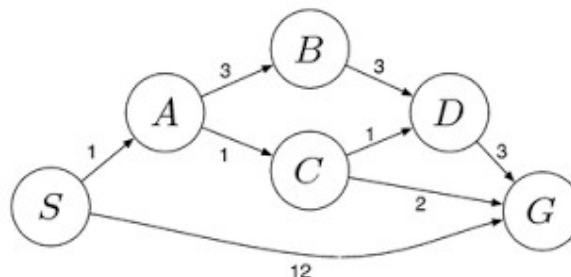
Karanam Bhaskar Gnanesh
Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Q1. Uniform Cost Search (UCS)

Problem

A logistics company wants to transport packages from **Warehouse S (Start)** to **Warehouse G (Goal)**.

Each road has a **cost**:

- Fuel cost
- Time cost

The company wants the **least-cost path**.

Why Uniform Cost Search?

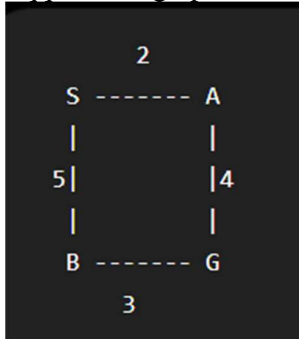
Uniform Cost Search always chooses the path with the **smallest total cost**.

It does **not** care about the fewest number of roads.

It only cares about the **minimum total cost**.

Example

Suppose the graph is:



Possible paths:

- $S \rightarrow A \rightarrow G$
 - $\text{Cost} = 2 + 4 = 6$
- $S \rightarrow B \rightarrow G$
 - $\text{Cost} = 5 + 3 = 8$

UCS selects:

$S \rightarrow A \rightarrow G$

because $6 < 8$.

How UCS Works

1. Start from **S**.
2. Put **S** into a priority queue with cost **0**.
3. Remove the node with the **lowest total cost**.
4. Expand its neighbors.
5. Update their costs.
6. Repeat until **G** is reached.
7. The first time **G** is removed from the queue, that path is the least-cost path.

Pseudocode

UCS(Start, Goal)

Create Priority Queue

Insert Start with cost = 0

WHILE Queue is not empty

 Remove node with minimum cost

 IF node == Goal

 Return Path and Cost

 FOR each neighbor

$\text{NewCost} = \text{CurrentCost} + \text{EdgeCost}$

IF neighbor not visited
 Insert neighbor with NewCost

END WHILE

Exam Answer (5 Marks)

Algorithm: Uniform Cost Search (UCS)

Reason: UCS always expands the node with the minimum cumulative path cost and guarantees the least-cost path.

Steps:

1. Start from source node **S**.
2. Insert **S** into a priority queue with cost **0**.
3. Expand the node with the smallest cost.
4. Add neighboring nodes with updated costs.
5. Repeat until destination **G** is reached.
6. Return the least-cost path.

Q2. Mars Rover – AI Agent

i. What are the percepts for this agent?

Percepts are the information the Mars Rover receives from its sensors.

Examples:

- Images of rocks and soil
- Surface temperature
- Obstacle detection
- Distance to objects
- Atmospheric conditions
- Battery level
- Position (GPS-like localization on Mars)

Answer:

The percepts are camera images, soil and rock data, temperature, obstacles, atmospheric information, battery status, and location.

ii. Characterize the operating environment.

The operating environment describes where the Mars Rover works.

Characteristics:

- **Partially observable** – The rover cannot see everything around it.
- **Deterministic/Stochastic** – Mostly uncertain because dust storms and terrain can change.
- **Sequential** – Each action affects future actions.
- **Dynamic** – The environment changes over time.
- **Continuous** – Movement and sensing happen continuously.
- **Single-agent** – Only one rover performs the task.

Answer:

The Mars Rover operates in a partially observable, uncertain, sequential, dynamic, continuous, and single-agent environment.

iii. What actions can the agent take?

The rover can:

- Move forward, backward, left, and right

- Avoid obstacles
- Collect rock and soil samples
- Drill rocks
- Capture photos and videos
- Analyze samples
- Send collected data to Earth
- Recharge using solar panels (if available)

Answer:

The rover can move, avoid obstacles, collect samples, analyze rocks, capture images, communicate with Earth, and manage its power.

iv. How can one evaluate the performance of the agent?

The rover's performance is measured by:

- Successfully collecting samples
- Accuracy of scientific analysis
- Safe navigation
- Number of obstacles avoided
- Battery efficiency
- Amount of useful data sent to Earth
- Completing the mission within the available time

Answer:

The agent is evaluated based on mission success, safe navigation, sample collection, scientific accuracy, energy efficiency, and successful communication.

v. What sort of agent architecture is most suitable? Why?

Suitable Architecture: Utility-Based Learning Agent

Why?

- Learns from previous experiences.
- Chooses the best action in uncertain situations.
- Balances multiple goals like safety, energy saving, and scientific exploration.
- Adapts to unknown environments on Mars.

Answer:

A **Utility-Based Learning Agent** is the most suitable because it can learn from experience, make intelligent decisions, adapt to changing conditions, and maximize mission success.

Q3. OLA Cab Booking Problem

Question

A customer wants to travel from one location to another using the OLA Cab app. The customer can choose a cab type (Mini, Micro, Sedan, Shared, Prime, etc.) based on comfort.

Identify the type of problem-solving agent and write the pseudocode.

1. Which type of problem-solving agent is used?

Answer: Goal-Based Problem-Solving Agent

Why?

The customer's **goal** is to **reach the destination**.

The agent:

- Takes the source and destination.
- Finds available cab types.
- Allows the user to choose a cab.
- Books the selected cab.

- Helps the customer reach the destination.

Since it works toward achieving a specific goal, it is called a **Goal-Based Agent**.

2. State Space

In AI, a problem-solving agent is explained using four components.

- **Initial State:** Customer opens the OLA app.
 - **Actions:** Enter pickup & destination, choose cab type, book cab.
 - **Goal State:** Customer reaches the destination.
 - **Path Cost:** Fare, travel time, or distance.
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3. Pseudocode

START

Input Pickup_Location

Input Destination

Display available cab types

(Mini, Micro, Sedan, Shared, Prime)

User selects cab type

IF cab is available THEN

 Show fare and estimated arrival time

 Confirm booking

 Assign nearest driver

 Driver picks up customer

 Customer reaches destination

 Display "Trip Completed"

ELSE

 Display "No Cab Available"

ENDIF

STOP

4. Simple Explanation

1. Customer opens the OLA app.
 2. Enters pickup and destination.
 3. App searches for nearby cabs.
 4. Customer selects a cab type (Mini, Sedan, Prime, etc.).
 5. If a cab is available, the booking is confirmed.
 6. The driver arrives and takes the customer to the destination.
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5. Exam Answer (5 Marks)

Type of Agent: Goal-Based Problem-Solving Agent

Reason: It aims to achieve the customer's goal of reaching the destination by selecting and booking the most suitable cab.

Pseudocode:

START

Input source and destination

Search available cabs

```
Display cab types
User selects cab
IF cab available
    Book cab
    Assign driver
    Travel to destination
    Trip completed
ELSE
    Show "No cab available"
END IF
STOP
```